

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester VI Examination, 2022-23

Paper No. MTB-604: Discrete Mathematics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. (Compulsory — 2 marks each.)

- (a) Construct the truth table of the logical expression:

$$(p \vee q) \wedge \neg(p \wedge q)$$

- (b) Rewrite the following statements without using the conditional operator (
- $\rightarrow$
-):

(i) If it is cold, he wears a hat.

(ii) If productivity increases, then wages rise.

- (c) How many reflexive relations are there on a set with
- n
- elements?

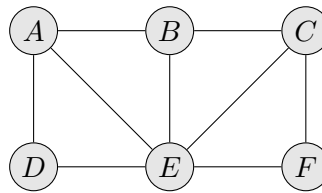
- (d) What is a totally ordered set (chain)? Present an example of a poset which is not totally ordered, and justify your answer.

- (e) Let
- $(L, \leq)$
- be a lattice. If
- $a \leq b$
- , then prove that for any
- $c \in L$
- :

$$a \wedge c \leq b \wedge c$$

- (f) Draw all possible trees with exactly 5 vertices.

- (g) What is an Euler circuit? Find an Eulerian path for the following graph, if it exists:

**Question 2.**

- (a) Let
- p
- be “It is cold” and let
- q
- be “It is raining”. Give a simple verbal sentence which describes each of the following statements: [6]

(i) $\neg p$ (ii) $p \wedge q$ (iii) $p \vee q$

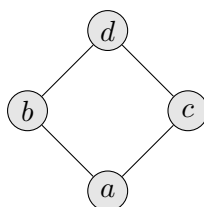
- (iv) $q \vee \neg p$
- (v) $p \rightarrow q$
- (vi) $p \leftrightarrow q$
- (b) Show that $\neg(p \vee q) \vee (\neg p \wedge q)$ and $\neg p$ are logically equivalent by developing a series of logical equivalences. [4]
- (c) When is a compound statement called: (i) a tautology, (ii) a contradiction, and (iii) a contingency? Give a simple example for each case. [4]
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Question 3.

- (a) Consider the conditional proposition $p \rightarrow q$. State its converse, inverse, and contrapositive propositions. Which of these propositions, if any, are logically equivalent to $p \rightarrow q$? Justify your answers. [6]
- (b) Negate each of the following statements: [4]
- (i) All students live in the dormitories.
- (ii) Some students are 25 years old or older.
- (c) Prove that if n is an integer and $3n + 2$ is odd, then n is odd. [4]
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Question 4.

- (a) Define a lattice. What are the distributive laws of a lattice? Give an example of a lattice in which the distributive laws (i) hold, and (ii) do not hold. Justify your answer. [6]
- (b) Draw the Hasse diagram representing the partial order relation $R = \{(a, b) \mid a \text{ divides } b\}$ on the set $S = \{2, 4, 5, 10, 12, 20, 25\}$. Find the maximal and minimal elements. [4]
- (c) Determine the poset whose Hasse diagram is given below. Is this poset a lattice? Justify your answer. [4]

**Question 5.**

- (a) Obtain the complete Conjunctive Normal Form (CNF) and Disjunctive Normal Form (DNF) of the Boolean function:

$$f(x, y, z, w) = xy + xzw$$

[6]

- (b) Use a Karnaugh map to find a minimal sum-of-products (SOP) form for the Boolean expression:

$$F(x, y, z, t) = xy't + x yzt + x'y'z' + x' yzt'$$

Moreover, draw the logic circuit corresponding to that minimal Boolean expression.

[6+2]

Question 6.

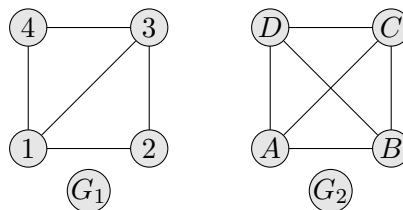
- (a) Let G be a connected planar simple graph with e edges and v vertices. Let r be the number of regions in a planar representation of G . Prove that:

$$r = e - v + 2$$

[6]

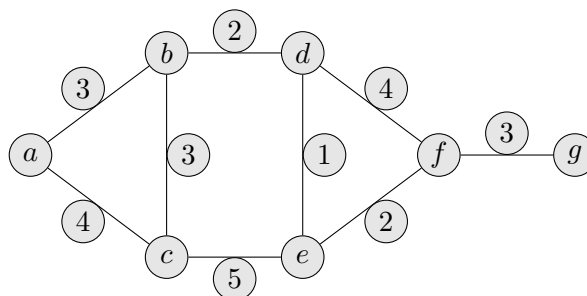
- (b) Prove or disprove that the complete graph K_5 and the complete bipartite graph $K_{3,3}$ are non-planar. [4]

- (c) When are two graphs said to be isomorphic? Determine, with justification, whether the following two graphs G_1 and G_2 are isomorphic: [4]



Question 7.

- (a) Let G be the following undirected weighted graph:



- (i) Use Kruskal's algorithm to find a Minimum Spanning Tree (MST) of G . [5]
- (ii) Use Dijkstra's shortest path algorithm to compute the shortest path from source vertex a to the destination vertex g of the graph G . [5]
- (b) Prove that any tree with at least two vertices has at least two pendant (leaf) vertices. [4]

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